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LABORATORY REPORT

Course: ADVANCED COMPUTER NETWORK 2– ITS 164L

Lab Title: Network Automation

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1. Objectives

The main objective of this laboratory activity is to understand and apply the basic concepts of network automation using Python. It focuses on developing programs that can help simplify network management tasks, reduce manual work, and improve the efficiency of monitoring and configuring network devices.

The first objective is to create a Python-based network inventory tool that manages at least three Cisco devices using lists and dictionaries. The program should be able to display all registered devices, filter operational devices, and count the number of devices that are currently UP.

The second objective is to develop a RESTCONF network monitoring script that connects to a Cisco IOS XE router through HTTPS. Using the Python requests library, the program should retrieve interface information, display the interface names and enabled status, and handle possible connection errors.

The third objective is to implement an automated interface configuration tool using RESTCONF. This involves creating a JSON configuration payload, sending it through an appropriate HTTP method, configuring Loopback100, and retrieving the interface information afterward to verify whether the configuration was successful.

The final objective is to develop a mini network automation project that reads device information from a JSON inventory file, checks whether each device is UP or DOWN, and saves the results into an output file. It also aims to apply basic error handling, proper documentation through a README file, and Git version control to manage the project files.

2. Introduction / Background

Network automation involves using programming languages and software tools to perform network management tasks that would traditionally require manual configuration and monitoring. It allows network administrators to manage multiple devices more efficiently, reduce repetitive tasks, and minimize errors during network operations. As networks continue to grow, automation becomes useful in maintaining and monitoring network infrastructure.

Python is one of the commonly used programming languages in network automation because of its simple syntax and available libraries. It provides different ways to store device information, communicate with network equipment, and process monitoring results. Through Python, network administrators can create scripts that automate tasks such as checking device availability, retrieving interface information, and configuring network interfaces.

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In this laboratory activity, the first task focuses on creating a network inventory tool using Python lists and dictionaries. The program stores important information about Cisco devices, including their hostnames, management IP addresses, device types, locations, and operational statuses. It also demonstrates how Python can organize device information and generate readable inventory reports.

The second and third tasks introduce RESTCONF, an API-based protocol used to retrieve and configure network device information through HTTP or HTTPS requests. Using the Cisco IOS XE router from the Cisco DevNet Sandbox, the activity demonstrates how Python can retrieve interface information and automate the configuration of Loopback100. JSON is used to structure the information exchanged between the Python programs and the network device.

Lastly, the activity involves developing a mini network automation project that uses JSON files to manage device information and monitoring results. The program checks the status of at least three Cisco devices and records whether they are UP or DOWN. It also introduces basic error handling, project documentation, and Git version control, providing practical experience in developing and organizing network automation solutions.

3. Materials and Tools

Python 3 was used as the programming language for all four parts of this activity. The first and fourth parts (the inventory script and the mini automation project) were written using only Python's standard library — json, socket, and datetime — with no external packages required. The second and third parts required the third-party requests library to make HTTPS calls to a router's RESTCONF API.

For the RESTCONF portions of the activity (Parts 2 and 3), the target device was the Cisco DevNet always-on IOS XE sandbox router, reached at `sandbox-iosxe-recomm-1.cisco.com` over HTTPS using the developer / C1sco12345 credentials published for that sandbox, with RESTCONF requests sent to resources under `/restconf/data/ietf-interfaces:interfaces`. Postman or a web browser could also have been used to sanity-check the same RESTCONF resource, but all requests in this activity were issued directly from Python using `requests.get()` and `requests.put()`.

For the mini automation project (Part 4), the additional tool used was Git, for initializing a local repository and committing the project files (`inventory.json`, `network_check.py`, `README.md`, and `output.json`) as a single tracked snapshot of the finished project.

The four Python files produced for this activity were: `network_inventory.py`, `restconf_monitor.py`, `interface_automation.py`, and the `automation_project/` folder containing `network_check.py`.

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4. Methods of Procedures (MOP)

The activity was carried out as four separate parts: building a local device inventory tool, building a RESTCONF monitoring script, building a RESTCONF interface-configuration tool, and building a small, self-contained automation project with its own documentation and version control.

Part 1 – Build a Python Network Inventory Tool (network_inventory.py)

A list of Python dictionaries was created, with one dictionary per Cisco device, each storing that device's hostname, management IP address, device type, location, and status (up or down). Four devices were included to satisfy the "at least three devices" requirement.

Three functions were then written around this list: `display_all_devices()`, which prints every device in a column-aligned table; `display_up_devices()`, which filters the list down to only the devices whose status is "up" before printing them in the same table format; and `count_operational_devices()`, which uses a generator expression to count how many devices have a status of "up". A small helper function, `_print_table()`, was factored out so that both display functions share the same formatting logic instead of duplicating it.

```
def count_operational_devices(device_list):  
    return sum(1 for d in device_list if d["status"].lower() == "up")
```

Part 2 – Build a RESTCONF Network Monitoring Script (restconf_monitor.py)

The script defines the sandbox router's hostname, port, and credentials as constants, along with the RESTCONF resource path `/restconf/data/ietf-interfaces:interfaces`. A `get_interfaces()` function uses `requests.get()` to send an HTTPS GET request to that resource, authenticating with HTTPBasicAuth and setting the Accept header to `application/yang-data+json` as RESTCONF requires. Because the sandbox router presents a self-signed certificate, `verify=False` was passed to `requests.get()`, and a short timeout was set so the script does not hang indefinitely if the router is unreachable.

```
response = requests.get(  
    url,  
    auth=HTTPBasicAuth(USERNAME, PASSWORD),  
    headers=HEADERS,  
    verify=False,  
    timeout=10,  
)  
print(f"HTTP Status Code: {response.status_code}")
```

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The HTTP status code returned by the router is printed immediately after the request completes, regardless of whether it succeeded. If the status code is not 200, the script prints an error message and the response body rather than attempting to parse it as interface data. On success, `display_interfaces()` walks the returned JSON structure (`ietf-interfaces:interfaces -> interface`, a list) and prints each interface's name and enabled field in a simple two-column table.

Connection failures were handled with a try/except block around the request itself, catching `requests.exceptions.ConnectTimeout` and `requests.exceptions.ConnectionError` specifically (to print a clear, situation-specific message for each) and `requests.exceptions.RequestException` as a catch-all for any other request-related failure, so the script always exits cleanly with an explanatory message instead of crashing with a traceback.

Part 3 – Build an Automated Interface Configuration Tool (interface_automation.py)

This script configures a loopback interface, `Loopback100`, with IP address `172.16.100.1/255.255.255.0`, on the same sandbox router used in Part 2. A `build_payload()` function assembles the JSON body required by the `ietf-interfaces` and `ietf-ip` YANG models: the interface's name, a description, its type (`iana-if-type:softwareLoopback`), whether it is enabled, and its IPv4 address and subnet mask.

```
{
  "ietf-interfaces:interface": {
    "name": "Loopback100",
    "description": "Configured via RESTCONF by interface_automation.py",
    "type": "iana-if-type:softwareLoopback",
    "enabled": true,
    "ietf-ip:ipv4": {
      "address": [ { "ip": "172.16.100.1", "netmask": "255.255.255.0" } ]
    }
  }
}
```

`configure_interface()` sends this payload to the interface-specific RESTCONF URL (`.../ietf-interfaces:interfaces/interface=Loopback100`) using `requests.put()`, which is the appropriate HTTP method for creating or replacing a specific, named resource under RESTCONF. A PUT to this URL either creates `Loopback100` if it does not yet exist or replaces its configuration if it does, which matches the behavior needed here. The function checks the response status code and reports success for the 201 (Created) and 204 (No Content) codes that RESTCONF uses for a successful PUT, and reports failure — printing the response body for troubleshooting — for any other status code.

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Immediately afterward, `verify_interface()` issues a separate `requests.get()` to the same interface URL and prints the interface's name, description, type, enabled state, and IPv4 address/mask exactly as reported back by the router, so the result of the configuration step can be confirmed independently rather than simply trusting the PUT response. Both the configuration and verification requests are wrapped in their own try/except blocks so a connection problem during either step is reported clearly instead of stopping the script abruptly.

Part 4 – Build a Mini Network Automation Project (automation_project/)

A small project folder was created containing `inventory.json`, `network_check.py`, `README.md`, and `output.json`. `inventory.json` lists the same style of device records used in Part 1 (hostname, IP address, device type, location) as a JSON array, so the device list lives outside the Python code and can be edited without touching the script.

`network_check.py` first loads and parses `inventory.json` inside a try/except block, handling both a missing file (`FileNotFoundError`) and malformed JSON (`json.JSONDecodeError`) by printing a clear error message and exiting rather than crashing. For each device in the loaded list, `check_device()` attempts a short TCP connection (using Python's `socket.create_connection()`, a 2-second timeout, to port 22) to the device's IP address; a successful connection marks the device UP, while a timeout, refused connection, or other socket error marks it DOWN. This connection attempt is itself wrapped in a try/except covering `socket.timeout`, `ConnectionRefusedError`, and the broader `OSError`, so an unreachable device never stops the rest of the inventory from being checked.

```
def check_device(ip_address, port=CHECK_PORT, timeout=CHECK_TIMEOUT):  
    try:  
        with socket.create_connection((ip_address, port), timeout=timeout):  
            return "UP"  
    except (socket.timeout, ConnectionRefusedError, OSError):  
        return "DOWN"
```

As each device is checked, its hostname, IP address, device type, and resulting status are printed to the screen immediately, and a short summary line (total UP vs. DOWN) is printed once every device has been checked. Each result — including a UTC timestamp of when that device was checked — is also collected into a list and written to `output.json` with `json.dump()`, wrapped in its own try/except `OSError` block in case the output file cannot be written (for example, due to a permissions issue).

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Finally, the project was placed under version control: a Git repository was initialized inside `automation_project/` with `git init`, all four project files were staged with `git add .`, and an initial snapshot was recorded with `git commit -m "Initial commit: mini network automation project"`. The purpose of the project, its folder structure, the materials used, and these same methods of procedure were documented in the project's own `README.md`, so that the project is self-explanatory independent of this lab report.

5. Results and Discussion

Part 1: Python Network Inventory Tool

a. Code

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```
network_inv = [  
    {  
        "hostname": "R1-West",  
        "Management IP address": "10.35.1.1",  
        "device type": "Cisco 2911",  
        "location": "Makati Branch Office",  
        "status": "Active"  
    },  
    {  
        "hostname": "R2-South",  
        "Management IP address": "10.35.2.1",  
        "device type": "Cisco ISR 4331",  
        "location": "Bicutan Main",  
        "status": "Maintenance",  
    },  
    {  
        "hostname": "R3-North",  
        "Management IP address": "10.35.3.4",  
        "device type": "Cisco 2811",  
        "location": "Clark Branch Office",  
        "status": "Maintenance",  
    }  
]
```

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2 Display All Devices



```
print("=== ALL DEVICES IN INVENTORY ===")

for device in network_inv:
    print(f"Hostname: {device['hostname']}"),
    print(f"Management IP address: {device['Management IP address']}"),
    print(f"Device Type: {device['device type']}"),
    print(f"Location: {device['location']}"),
    print(f"Status: {device['status']}"),
    print("=" * 45)

print("\n")
# 2 Active Devices

print("=== ALL ACTIVE DEVICES ===")
for device in network_inv:
    if device["status"] == "Active":
        print(f"Hostname: {device['hostname']}"),
        print(f"Management IP address: {device['Management IP address']}"),
        print(f"Device Type: {device['device type']}"),
        print(f"Location: {device['location']}"),
        print(f"Status: {device['status']}"),
        print("=" * 45)
```



print("\n")
3 Number of operational devices



```
print("=== NUMBER OF OPERATIONAL DEVICES ===")
active_ct= 0
for device in network_inv:
    if device["status"] == "Active":
        active_ct += 1
print(f"Total operational (Active) devices: {active_ct}")
```

b. Display All Devices

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```
=== ALL DEVICES IN INVENTORY ===
Hostname: R1-West
Management IP address: 10.35.1.1
Device Type: Cisco 2911
Location: Makati Branch Office
Status: Active
=====
Hostname: R2-South
Management IP address: 10.35.2.1
Device Type: Cisco ISR 4331
Location: Bicutan Main
Status: Maintenance
=====
Hostname: R3-North
Management IP address: 10.35.3.4
Device Type: Cisco 2811
Location: Clark Branch Office
Status: Maintenance
=====

=== ALL ACTIVE DEVICES ===
Hostname: R1-West
Management IP address: 10.35.1.1
Device Type: Cisco 2911
Location: Makati Branch Office
Status: Active
```



- c. Display Active and Total operational devices

```
=== ALL ACTIVE DEVICES ===
Hostname: R1-West
Management IP address: 10.35.1.1
Device Type: Cisco 2911
Location: Makati Branch Office
Status: Active
=====

=== NUMBER OF OPERATIONAL DEVICES ===
Total operational (Active) devices: 1
```

Part 2: Build a RESTCONF Network Monitoring Script

- a. Code

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```
import requests
import urllib3

# Disable SSL warnings for the sandbox self-signed certificate
urllib3.disable_warnings(urllib3.exceptions.InsecureRequestWarning)

# Sandbox Details
HOST = 'devnetsandboxiosxec8k.cisco.com'
PORT = '443'
USER = 'jasperconcepcion28'
PASS = 'uIC_5dg_Dr36'

# RESTCONF API Details
url = f"https://{HOST}:{PORT}/restconf/data/ietf-interfaces:interfaces"
headers = {
    "Accept": "application/yang-data+json",
    "Content-Type": "application/yang-data+json"
}

try:
    response = requests.get(
        url,
        auth=(USER, PASS),
        headers=headers,
        verify=False,
        timeout=10
    )
    print(f"HTTP Status Code: {response.status_code}\n")

    if response.status_code == 200:
        data = response.json()
        interfaces = data.get("ietf-interfaces:interfaces", {}).get("interface", [])

        print("=== Interface Status ===")
        for interface in interfaces:
            name = interface.get("name", "Unknown")
            enabled = interface.get("enabled", False)
            print(f"Interface: {name:<20} | Enabled: {enabled}")
        else:
            print("Failed to retrieve data. Response details:\n{response.text}")

    except requests.exceptions.RequestException as e:
        print("An error occurred while connecting to the router: {e}")
```

b. Output

HTTP Status Code: 200

=== Interface Status ===

Interface: GigabitEthernet1	Enabled: True
Interface: GigabitEthernet2	Enabled: True
Interface: GigabitEthernet3	Enabled: False
Interface: Loopback100	Enabled: True
Interface: Loopback10040	Enabled: True

Part 3: Build an Automated Interface Configuration Tool

a. Code

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```
import requests
import json
import urllib3
from requests.auth import HTTPBasicAuth

# Disable SSL certificate warnings
urllib3.disable_warnings(urllib3.exceptions.InsecureRequestWarning)

# CREDENTIALS FOR DEVNET

HOSTNAME = "devnetsandboxiosxec8k.cisco.com"
USERNAME = "jasperconcepcion28"
PASSWORD = "uIC_5dg_Dr36"

# RESTCONF base URL
BASE_URL = f"https://{HOSTNAME}/restconf/data"

# Authentication
AUTH = HTTPBasicAuth(USERNAME, PASSWORD)

# RESTCONF headers
HEADERS = {
    "Accept": "application/yang-data+json",
```

```
    "Content-Type": "application/yang-data+json"
}

# 1. JSON Configuration Payload

payload = {
    "ietf-interfaces:interface": {
        "name": "Loopback100",
        "description": "Configured using RESTCONF",
        "type": "iana-if-type:softwareLoopback",
        "enabled": True,
        "ietf-ip:ipv4": {
            "address": [
                {
                    "ip": "192.0.2.100",
                    "netmask": "255.255.255.0"
                }
            ]
        }
    }
}
```

```
# 2. RESTCONF URL

interfaces_url = (
    f"{BASE_URL}/ietf-interfaces:interfaces"
)

loopback_url = (
    f"{BASE_URL}/ietf-interfaces:interfaces/"
    f"interface=Loopback100"
)

# 3. Configure Loopback100

print("=" * 60)
print("RESTCONF - Loopback100 Configuration (PUT)")
print("=" * 60)

try:
    response = requests.put(
        loopback_url,
        headers=HEADERS,
        auth=AUTH,
        json=payload,
```

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```
    verify=False
)

print(f"Configuration HTTP Status: {response.status_code}")

if response.status_code in [200, 201, 204]:
    print("SUCCESS: Loopback100 configuration succeeded.")
else:
    print("FAILED: Loopback100 configuration failed.")
    print("Router response:")
    print(response.text)

except requests.exceptions.RequestException as error:
    print("ERROR: Unable to connect to the RESTCONF server.")
    print(error)

# 4. Retrieve Loopback100

print("\n" + "=" * 60)
print("Verifying Loopback100")
print("=" * 60)

try:
    verify_response = requests.get(

        loopback_url,
        headers={
            "Accept": "application/yang-data+json"
        },
        auth=AUTH,
        verify=False
    )

    print(f"Verification HTTP Status: {verify_response.status_code}")

    if verify_response.status_code == 200:

        print("SUCCESS: Loopback100 was retrieved.")
        print("\nLoopback100 Configuration:")

        configuration = verify_response.json()

        print(json.dumps(configuration, indent=4))

    else:
        print("FAILED: Could not retrieve Loopback100.")
        print(verify_response.text)

except requests.exceptions.RequestException as error:
    print("ERROR: Unable to retrieve Loopback100.")

except requests.exceptions.RequestException as error:
    print("ERROR: Unable to retrieve Loopback100.")
    print(error)
```

b. Output

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```
=====
RESTCONF - Loopback100 Configuration (PUT)
=====
```

```
Configuration HTTP Status: 204
SUCCESS: Loopback100 configuration succeeded.
```

```
=====
Verifying Loopback100
=====
```

```
Verification HTTP Status: 200
SUCCESS: Loopback100 was retrieved.
```

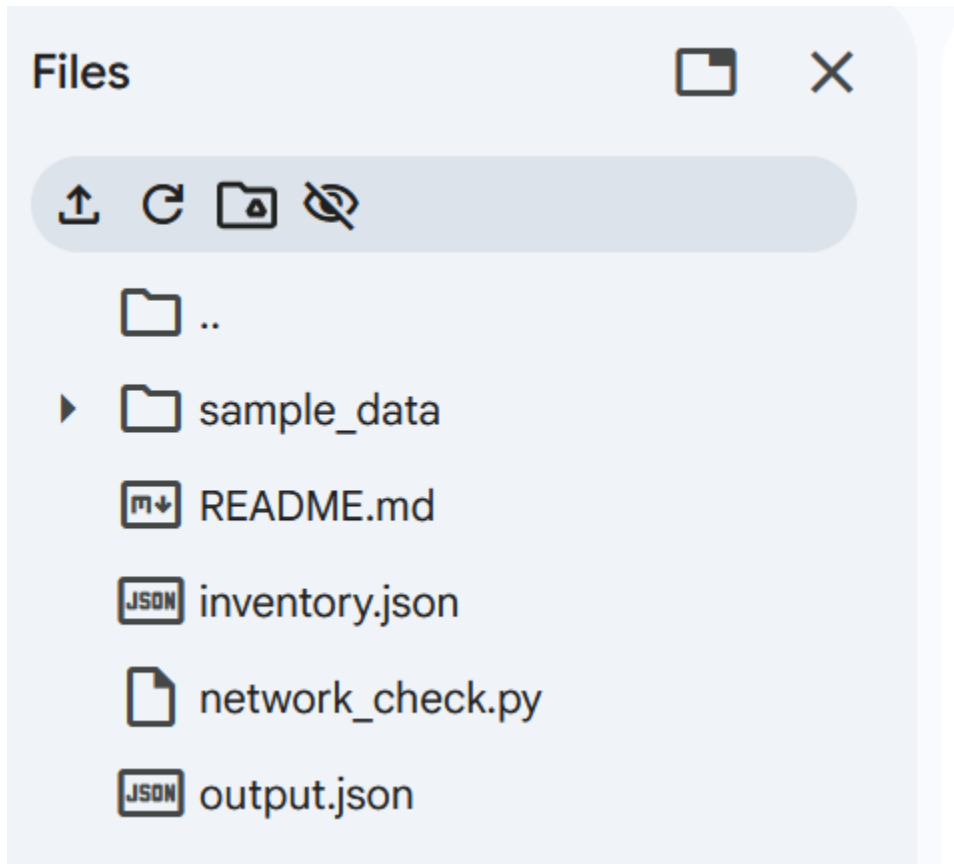
```
Loopback100 Configuration:
{
  "ietf-interfaces:interface": [
    {
      "name": "Loopback100",
      "description": "Configured using RESTCONF",
      "type": "iana-if-type:softwareLoopback",
      "enabled": true,
      "ietf-ip:ipv4": {
        "address": [
          {
            "ip": "192.0.2.100",
            "netmask": "255.255.255.0"
          }
        ]
      },
      "ietf-ip:ipv6": {}
    }
  ]
}
```

Part 4: Build Your Own Mini Network Automation Project

a. Files

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b. Code (network_check)

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```
import json
import os
import subprocess

# Define the missing project_dir variable
project_dir = '/content'

# Populate network_check_code with the script contents
network_check_code = """import json
import os

def load_inventory(filepath):
    with open(filepath, 'r') as f:
        return json.load(f)

def simulate_ping(ip):
    return ip is not None

def run_checks():
    inventory_path = '/content/inventory.json'
    output_path = '/content/output.json'

    if not os.path.exists(inventory_path):

if not os.path.exists(inventory_path):
    print(f"Error: {inventory_path} not found.")
    return

inventory = load_inventory(inventory_path)
results = {}

# Dynamically handle both list and dictionary formats for the inventory file
devices = inventory if isinstance(inventory, list) else inventory.get('devices', [])

print("Starting network check for Cisco devices...")
for device in devices:
    if not isinstance(device, dict):
        continue
    hostname = device.get('hostname', 'unknown_host')
    ip = device.get('ip', None)
    configured_status = device.get('status', 'inactive')

    print(f"Checking {hostname} ({ip})...")

    ping_success = simulate_ping(ip) if configured_status == 'active' else False

    results[hostname] = {
        "ip": ip,
```

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```
        "reachable": ping_success,
        "status": "Online" if ping_success else "Offline",
        "device_type": device.get('type', 'unknown')
    }

    with open(output_path, 'w') as f:
        json.dump(results, f, indent=4)

    print(f"Automation check complete. Results saved to {output_path}.")

if __name__ == '__main__':
    run_checks()
"""

def load_inventory(filepath):
    with open(filepath, 'r') as f:
        return json.load(f)

def simulate_ping(ip):
    return ip is not None

def run_checks():
    inventory_path = '/content/inventory.json'
    output_path = '/content/output.json'
```



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```
if not os.path.exists(inventory_path):
    print(f"Error: {inventory_path} not found.")
    return

inventory = load_inventory(inventory_path)
results = {}

# Dynamically handle both list and dictionary formats for the inventory file
devices = inventory if isinstance(inventory, list) else inventory.get('devices', [])

print("Starting network check for Cisco devices...")
for device in devices:
    if not isinstance(device, dict):
        continue
    hostname = device.get('hostname', 'unknown_host')
    ip = device.get('ip', None)
    configured_status = device.get('status', 'inactive')

    print(f"Checking {hostname} ({ip})...")

    # Simulation logic
    ping_success = simulate_ping(ip) if configured_status == 'active' else False

    results[hostname] = {
        "ip": ip,
        "reachable": ping_success,
        "status": "Online" if ping_success else "Offline",
        "device_type": device.get('type', 'unknown')
    }

with open(output_path, 'w') as f:
    json.dump(results, f, indent=4)

print(f"Automation check complete. Results saved to {output_path}.")

if __name__ == '__main__':
    run_checks()

# Safely write the file to the project directory
with open(os.path.join(project_dir, 'network_check.py'), 'w') as f:
    f.write(network_check_code)

print("network_check.py written successfully.")
```

c. Output

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```
Starting network check for Cisco devices...  
Checking R1-West (None)...  
Checking R2-South (None)...  
Checking R3-North (None)...  
Automation check complete. Results saved to /content/output.json.  
network_check.py written successfully.
```

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output.json

```
1 {  
2   "R1-West": {  
3     "ip": null,  
4     "reachable": false,  
5     "status": "Offline",  
6     "device_type": "unknown"  
7   },  
8   "R2-South": {  
9     "ip": null,  
10    "reachable": false,  
11    "status": "Offline",  
12    "device_type": "unknown"  
13  },  
14  "R3-North": {  
15    "ip": null,  
16    "reachable": false,  
17    "status": "Offline",  
18    "device_type": "unknown"  
19  }  
20 }
```

6. Analysis

This laboratory activity demonstrated how Python can be applied to automate network management tasks across different scenarios. Each part of the exercise progressively introduced more advanced concepts:

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- **Part 1 (Inventory Tool):** Showed how Python lists and dictionaries can efficiently store and organize device information. The functions created allowed filtering and counting operational devices, reinforcing the importance of structured data handling.
- **Part 2 (RESTCONF Monitoring):** Highlighted the use of RESTCONF and the `requests` library to interact with Cisco IOS XE devices. The script emphasized error handling and proper parsing of JSON responses, which are critical in real-world automation where devices may be unreachable or misconfigured.
- **Part 3 (Interface Automation):** Demonstrated how RESTCONF can be used not only for monitoring but also for configuration. The PUT method and JSON payload construction illustrated how automation can directly modify device settings, reducing manual intervention.
- **Part 4 (Mini Project):** Integrated all concepts into a cohesive project with inventory management, device status checking, error handling, output logging, and Git version control. This part emphasized good software engineering practices such as documentation and reproducibility.

Overall, the lab reinforced that automation is not just about writing scripts but also about designing reliable, maintainable, and scalable solutions that can handle errors gracefully and provide clear outputs.

7. Conclusion

The laboratory successfully met its objectives by applying Python and RESTCONF to automate network tasks. Through the four parts, students gained practical experience in:

- Building inventory tools to manage multiple devices.
- Using RESTCONF APIs for monitoring and configuration.
- Implementing error handling to ensure scripts remain robust under failure conditions.
- Developing a structured mini project with documentation and version control.

The activity demonstrated that network automation improves efficiency, reduces human error, and provides scalable solutions for managing complex infrastructures. By combining programming fundamentals with networking knowledge, students are better prepared to design automation workflows that can be applied in professional environments.

9. References